

STACK

Q7)

Question:

DS	Session	Stack	Question Information	Level 1	Challenge 47
Question description You are given an array A of N integers. Now, two functions $F(X)$ and $G(X)$ are defined: $F(X)$: This is the smallest number Z such that $X < Z \leq N$ and $A[X] < A[Z]$ $G(X)$: This is the smallest number Z such that $X < Z \leq N$ and $A[X] > A[Z]$ Now, you need to find for each index i of this array $G(F(i))$, where $1 \leq i \leq N$. If such a number does not exist, for particular index i, output 1 as its answer. If such a number does exist, output $A[G(F(i))]$ Constraints: $1 \leq N \leq 3000$ $0 \leq A[i] \leq 10^{18}$ Input : The first line contains a single integer N denoting the size of array A. Each of the next N lines contains a single integer, where the integer on the i^{th} line denotes $A[i]$. Output : Print N space-separated integers on a single line, where the i^{th} integer denotes $A[G(F(i))]$ or 1, if $G(F(i))$ does not exist.					

PROGRAM:

```
1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      long long a[n];
8      int f[n], g[n], stack[n];
9      int top = -1;
10
11     for (int i = 0; i < n; i++)
12         scanf("%lld", &a[i]);
13
14     for (int i = n - 1; i >= 0; i--) {
15         while (top != -1 && a[stack[top]] <= a[i])
16             top--;
17
18         f[i] = (top == -1) ? -1 : stack[top];
19         stack[++top] = i;
20     }
21
22     top = -1;
23
24     for (int i = n - 1; i >= 0; i--) {
25         while (top != -1 && a[stack[top]] >= a[i])
26             top--;
27
28         g[i] = (top == -1) ? -1 : stack[top];
29         stack[++top] = i;
30     }
31
32     for (int i = 0; i < n; i++) {
33         if (f[i] == -1 || g[f[i]] == -1)
34             printf("-1 ");
35         else
36             printf("%lld ", a[g[f[i]]]);
37     }
38
39     return 0;
40 }
```

OUTPUT:

TEST CASE 1:

```
8
3 7 1 7 8 4 5 2
1 4 4 4 -1 2 -1 -1
```

TEST CASE 2:

```
8
31 71 11 17 81 14 51 21
11 14 14 14 -1 21 -1 -1
```